Although the title of Avellaneda and Stoikov's article is called *High-frequency trading in a limit order book*, they do not incorporate any LOB features such as price tick, price-time priority, order size, fee and rebate into their model. The only LOB related claim is that the exponential form of arrival intensity λ is coming from empirical studies of LOB.

A.4 Guilbaud and Pham (2013)

Guilbaud and Pham [7] introduce a number of LOB features not seen in the AS framework, and the formulation of their control problem is also significantly different. First, the prices of market maker's limit orders (S_t^a, S_t^b) are no longer continuous decision variables, but are either pegged to the best bid/ask[40] or *one tick better*. When the bid-ask spread is only one tick, a one-tick-better limit order means market order. Second, the mid price S_t^m is extended to jump diffusion (Lévy process)[41] and the discrete bid-ask spread S_t is modeled by an independent continuous-time Markov chain. Besides, market maker can choose the size of limit orders L_t^a, L_t^b as well as the time τ_n and size ζ_n of market orders, which are used to remove excess inventory. Lastly, the final liquidation value includes the cost of crossing the spread $(|Q_T|S_T/2)$ and a non-proportional exchange fee η.

$$\max_{S_t^a, S_t^b, L_t^a, L_t^b, \tau_n, \zeta_n} \mathbb{E}\Big(U\left(B_T + Q_T S_T^m - |Q_T|S_T/2 - \eta\right)\Big) \tag{104}$$

$$B_t = \int_0^t S_s^a L_s^a dN_s^a - \int_0^t S_s^b L_s^a dN_s^b - \sum_{\tau_n \le t} \left(\zeta_n S_{\tau_n}^m + |\zeta_n| S_{\tau_n}/2 + \eta\right) \tag{105}$$

$$Q_t = \int_0^t L_s^b dN_s^b - \int_0^t L_s^a dN_s^a + \sum_{\tau_n \le t} \zeta_n \tag{106}$$

$$S_t^m = \mu t + \sigma W_t + \int_{(0,t]\times(-1,1)} \xi \widetilde{N}^m(ds \times d\xi) + \int_{(0,t]\times(-1,1)^c} \xi N^m(ds \times d\xi) \tag{107}$$

$$S_t \sim \text{ independent continuous-time Markov chain} \tag{108}$$

where $\mu, \sigma > 0$ are constants.

References

[1] M. Avellaneda and S. Stoikov. High-frequency trading in a limit order book. *Quantitative Finance*, 8(3):217–224, 2008.

[2] L. Glosten and P. Milgrom. Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics*, 14(1):71–100, 1985.

[3] A. S. Kyle. Continuous auctions and insider trading. *Econometrica*, 53(6):1315, 1985.

[4] T. Ho and H. Stoll. Optimal dealer pricing under transactions and return uncertainty. *Journal of Financial Economics*, 9(1):47–73, 1981.

[40] The best bid and ask prices are still not in the price grid as the mid price is continuous.

[41] ξ in equation (107) is the continuous jump size of Lévy process.